CL and NLP

Assignment - 1

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1. $x_t = 0.5$

$h_{t-1} = 0.2$

$C_{t-1} = 0.4$

LSTM eq^s are :

$$f_t = \sigma(W_f x_t + U_f h_{t-1} + b_f)$$

$$i_t = \sigma(W_i x_t + U_i h_{t-1} + b_i)$$

$$\tilde{C}_t = \tanh(W_c \cdot x_t + U_c \cdot h_{t-1} + b_c)$$

$$C_t = f_t \cdot C_{t-1} + i_t \cdot \tilde{C}_t$$

$$o_t = \sigma(W_o \cdot x_t + U_o \cdot h_{t-1} + b_o)$$

$$h_t = o_t \cdot \tanh(C_t)$$

a) forget gate :

$$z_f = (0.7)(0.5) + (0.3)(0.2) + 0.1$$
$$= 0.5100$$

$$f_t = \sigma(z_f)$$
$$= \frac{1}{1+e^{-0.5100}} = \boxed{0.6248}$$

b) Input gate :

$$z_i = (0.6)(0.5) + (0.4)(0.2) + 0$$
$$= 0.3800$$

$$i_t = \frac{1}{1 + e^{-0.3800}}$$
$$= \boxed{0.5939}$$

c) candidate cell state $(\tilde{C}_t)$:

$$z_c = (0.9)(0.5) + (0.2)(0.2)$$
$$= 0.4900$$

$$\tilde{C}_t = \tanh(0.4900)$$
$$= \boxed{0.4542}$$

d) new cell state (C_t) :

$$C_t = f_t \cdot C_{t-1} + i_t \cdot \tilde{C}_t$$

$$C_t = (0.6248)(0.4) + (0.5939)(0.4542)$$
$$= \boxed{0.5197}$$

e) Output gate (O_t) :

$$z_o = (0.5)(0.5) + (0.5)(0.2) + 0.1$$
$$= 0.4500$$

$$O_t = \frac{1}{1 + e^{-0.4500}}$$
$$= \boxed{0.6106}$$

b) New hidden state (h_t):

$$h_t = o_t \cdot \tanh(c_t)$$

$$\therefore \tanh(0.5197) \approx 0.4775$$

$$h_t = (0.6106)(0.4775)$$
$$\approx \boxed{0.2915}$$

2. $D = 10$

$H = 32$

no. of gates = 4

a) Parameters $= 4(DH + H^2 + H)$ [∴ one bias vector per gate]

$$P = 4((10)\cdot(32) + (32)(32) + 32)$$
$$\boxed{P = 5504 \text{ parameters}}$$

b) $P = 4(DH + H^2 + 2H)$ [∴ Two bias vectors per gate].

$$P = 4[(10)\cdot(32) + (32)(32) + 2(32)]$$
$$\boxed{P = 5632 \text{ parameters}}$$

3.

$$\boxed{\text{Scaling factor} = f^k}$$

a) $f = 0.9$

After 10 steps:

$$f^{10} = (0.9)^{10}$$
$$= \boxed{0.3487}$$

After 50 steps:

$$f^{50} = (0.9)^{50}$$
$$= \boxed{0.0052}$$

b) for $k = 50$;

when $f = 0.5$;

$$f^{50} = (0.5)^{50}$$
$$\boxed{f^{50} = 8.8818 \times 10^{-16}}$$

when $f = 0.99$;

$$f^{50} = (0.99)^{50}$$
$$\boxed{f^{50} = 0.6050}$$

c) When the forget gate is 0.5, the gradient becomes almost zero after 50 time steps, making it extremely difficult for the network to learn long-term dependencies. In contrast, when the forget gate is 0.99, approx 60.50 % of the gradient remains after 50 steps.

4. Given:

Duration: 3 years.
days per year: 365
Hours per day: 24
Input lookback: 168 hours
Forecast horizon: 24 hours.

Date :

Features per hour : 3
stride : 1 hour.
Batch size : 64.

a) Total no. of hourly records :

$N = 3 \times 365 \times 24$
$= \boxed{26280}$

b) For stride = 1 ;

$\text{Windows} = N - L - H + 1$ ← first valid window
$\text{windows} = 26280 - 168 - 24 + 1$
$= \boxed{26089}$

c) For batch of 64 windows :

$X : (64, 168, 3)$

Each target contains next 24 hourly load values.

$Y : (64, 24)$.

d) $N_{train} = 0.70 \times 26280$
$= \boxed{18396}$

$W_{train} = 18396 - 168 - 24 + 1$
$= \boxed{18205}$

$\text{Batches} = \left\lceil \frac{\text{Training windows}}{\text{Batch size}} \right\rceil$

$= \lceil 18205 / 64 \rceil$

$= \lceil 284.4531 \rceil$

$= 285$ mini-batches per epoch.

e) $P = 4(DH + H^2 + H)$

$P = 4\left((3)(64) + (64)^2 + 64\right)$

$= \boxed{17408}$

5.

a) Min-Max normalisation :

$$x' = \frac{x - x_{min}}{x_{max} - x_{min}}$$

$$\therefore x' = \frac{95 - 20}{120 - 20}$$

$$\boxed{x' = 0.75}$$

b)

$$\text{Prediction} = \begin{cases} \text{Failure}, & p \geq 0.5 \\ \text{No failure}, & p < 0.5 \end{cases}$$

$\therefore 0.83 \geq 0.5$

$\therefore$ **Prediction : Failure**

c) True positives (TP) = 30.

False positives (FP) = Predicted failures − TP
= 45 − 30
= 15

∴ FP = 15

False neg. (FN) = Actual failures − TP
= 40 − 30
= 10

~~True neg (TN) = 1000 − 40~~
~~= 960.~~

Actual non-failures : 1000 − 40
= 960.

TN = Actual non failures − FP
= 960 − 15
= 945

- Precision : $\frac{TP}{TP + FP}$
$= \frac{30}{30 + 15}$
$= 66.67\ \%$

- Recall : $\frac{TP}{TP + FN}$
$= \frac{30}{30 + 10}$
$= 75.\%$

- F1-score : $\frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$.

$$F_1\text{-score} = \frac{2 \times \frac{30}{45} \times \frac{30}{40}}{\frac{30}{45} + \frac{30}{40}}$$

$$= \boxed{70.59\%}$$

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

$$= \frac{30 + 945}{1000}$$

$$= \boxed{0.9750} \text{ or } \boxed{97.50\%}$$

d) → It should generally prioritise recall, because failing to detect a genuinely failing pump can cause equipment damage, unexpected downtime or safety risks.

→ Accuracy can be misleading because the dataset is imbalanced: only 40/1000 cycles result in failure. A model that predicts "no failure" for every cycle would still achieve 96% accuracy while detecting none of failures.
∴ Recall, together with precision & F1-score provides more useful assessment of failure detection.